

Airtech IT Bach -01

MERN Stack Learning Roadmap

1. Web fundamentals

- HTML: semantic tags, forms, tables
- CSS: box model, flexbox, grid
- Tailwind CSS
- Responsive design: media queries, mobile-first
- Git & GitHub: init, commit, push, pull request

2. JavaScript

- Variables, data types, operators
- Functions, arrays, objects
- ES6+: arrow functions, destructuring, spread operator
- Promises, async/await
- Fetch API ও JSON

3. React (Frontend)

- Components ও JSX
- Props ও state
- Hooks: useState, useEffect, useContext
- React Router
- Forms, validation, API data fetching
- DaisyUI ব্যবহার করে UI তৈরি

4. Node.js & Express (Backend)

- Node.js runtime & npm
- Express server setup
- Routing ও middleware
- REST API Create
- Error handling
- JWT authentication ও authorization

5. MongoDB

- Database, collection, document ধারণা
- CRUD operations
- Validation

- Aggregation basics

6. Full-stack project

- React frontend + Express backend + MongoDB database
- User authentication
- CRUD features
- Protected routes
- Deployment & live demo link

7. Next Js (Basic)

- Next Js frontend + Express backend + Mongoose +MongoDB database